How-To Library

Informatica

Configuring User Activity Detection in Data Privacy Management

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Abstract

You can monitor user access to sensitive data in Data Privacy Management. This article describes the steps to configure user activity monitoring using the Informatica AuditTrail Utility. It also describes the steps to import user information from a .csv file into Data Privacy Management.

Supported Versions

- Data Privacy Management 10.4.1

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Overview

Configure user activity monitoring in Data Privacy Management to monitor access to sensitive data in an organization.

You can use user activity monitoring to perform the following tasks:

- Understand which users have access to sensitive data and how users access the data.
- View summary data of activity events on the User Activity Summary workspace.
- Export user access information in a `.csv` file.
- To further analyze access patterns and spot behavior anomalies, you can configure security policies and anomaly detection.

You can use the Informatica Database AuditTrail Utility to configure activity streaming to Data Privacy Management. Event messages are streamed every hour.

To configure user activity monitoring in Data Privacy Management with the Informatica Database AuditTrail Utility, perform the following high-level tasks:

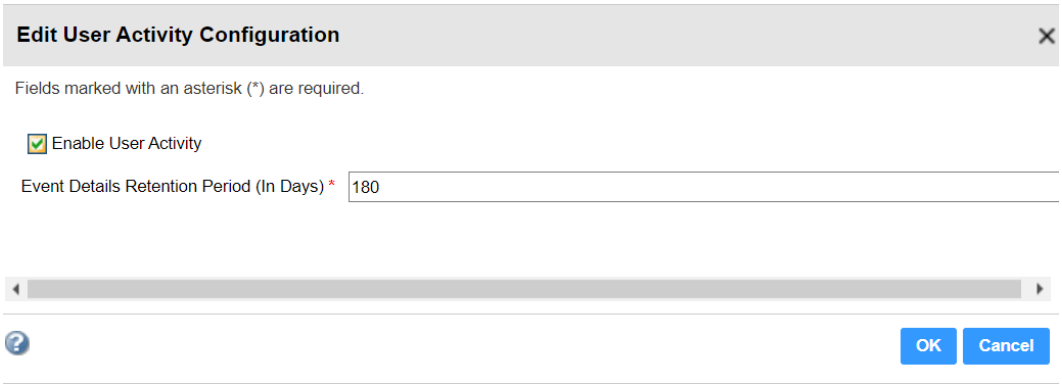
1. Enable User Activity detection on the Data Privacy Management Service.
2. Create a data store that connects to the database you want to monitor.
3. If required, import data store alias information.
   Required if the data store and database names are different.
4. Create and run a domain discovery scan to identify sensitive data.
5. Import user information from a `.csv` file.
6. Download the Informatica Database AuditTrail Utility and perform the required configuration to enable activity streaming.

Enable User Activity Detection on the Data Privacy Management Service

Before you can monitor user activity, you must configure the Data Privacy Management Service for user activity detection.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. Click the Data Privacy Management Service to open the properties page on the right.
4. On the **User Activity Configuration** tab, click the **Edit** button to open the **Edit User Activity Configuration** window.
5. Choose to enable user activity and enter the number of days to retain event data. Data is deleted at the end of the retention period.

   Default is 180 days.
   The following image shows the **Edit User Activity Configuration** window:



6. Click **OK**.
7. Recycle the Data Privacy Management Service.

To verify that user activity detection is activated, perform the following steps:

1. Log in to Data Privacy Management.
2. Click **Manage** > **Jobs** to open the **Jobs** workspace.
3. Click the **System Jobs** tab and verify that the following jobs are running:
   - Launch/Monitor TCP Listener

- Augmenter
- Percolator
- Elasticsearch
- UBAManager. Optional for User Activity.

You can also verify that the applications are running from the Hadoop Yarn Resource Manager.

Create a Data Store

Create a data store to connect to the database that you want to monitor. Configure the properties for the data source, and configure additional properties for reporting, grouping, and security.

For detailed property descriptions, see the Data Store Properties chapter in the *Informatica Data Privacy Management User Guide*.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. Click **New**.

   The **New_Data_Store** page appears.
3. Enter the connection properties for the data store.

   After you select the category and the data store type, additional connection properties appear.
4. Click **Test Connection** to verify that Data Privacy Management can connect to the data store with the connection details and log in.

   The connection test validates that the connection properties are configured, the connection is available, and the login credentials are valid.
5. Click **Save**.

   After you save a data store, you cannot edit the category or the data store type.

Import Data Store Alias Information

Data Privacy Management gathers user activity information from user actions on the source database. If a data store name differs from the original source name, you must give the data store an alias name. When you provide a data store alias name that matches the original source name, you link user activity information for that source to the corresponding data store.

For information about the format of the `.csv` file and rules and guidelines around import, see the Data Stores chapter in the *Informatica Data Privacy Management User Guide*.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. From the **Actions** menu, select **Import** > **Data Store Aliases**.

   The **Import Data Store Aliases** page appears.
3. Enable or disable the **Replace Duplicates with Items Imported** option.

   The option determines how the Import job handles updates for data store aliases that exist in the repository. If you enable the option, the Import job updates the data store alias properties in the repository with the values in the `.csv` file.

Warning: If a column in the `.csv` file contains a null value, the Import job replaces the value in the repository with a null value.

If you disable the option, the Import job skips the row.

4. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

5. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the `.csv` file, you can download a list of the rejected rows.

Create and Run a Domain Discovery Scan

Create a scan to configure how and when the scan job discovers and classifies sensitive data. You specify the type of scan, configure the options, and schedule when the scan runs. When you run the scan, Data Privacy Management creates a scan job and runs the job according to the scan schedule.

1. Click **Manage** > **Scans**.

   The **Scans** workspace appears.

2. Click **New**.

   The **Define & Connect** panel appears.

3. Enter a name and optionally, enter a description of the scan.
4. Choose the **Discover Domains** scan type.
5. Choose the category. Then select the data store type and the data stores to include in the scan.
6. Select the scan options.
7. Click **Next**.

   The **Assign Classification Policy** panel appears.

8. Select the classification policies to assign to the scan. You can use the following filters:

   - **Classification Policies Selected**. Displays a list of the classification policies that are assigned to the scan. Use the filter to view or remove classification policies.
   - **Assign Classification Policies**. Displays a list of all classification policies in the Data Privacy Management repository. Use the filter to add or remove classification policies.

9. Click **Next**.

   The **Schedule** panel appears.

10. Choose to run the scan immediately.
11. Click **Save**.

Import User Information

You can import different types of user metadata from multiple sources. The process that you follow depends on what you want to import and from which source. To import user metadata from multiple sources, follow the process for each source.

The following table lists the supported import types for each source:

Import Type	Supported LDAP Directory Services	Salesforce	.csv Files
Users	Yes	Yes	Yes
User groups	Yes	Yes	Yes
User group memberships	Yes	Yes	Yes
User access	No	Yes	Yes
User activity	No	Yes	No
User aliases	No	No	Yes

You can import data from a `.csv` file. When you import user metadata, you schedule an import job. The import job saves users, groups, group memberships, user access, and user aliases metadata in the Data Privacy Management repository.

If user information changes in the source system, you must update Data Privacy Management to reflect the changes. Use the same method to create and update user metadata. For example, if you imported users from a `.csv` file, import changes in a `.csv` file.

The `.csv` file that you use to import the user information must be in the required format. The required column headings differ based on the kind of information to be imported. Verify that the file contains all required columns and headings before you perform an import.

Rules and Guidelines to Import User Information from a .csv File

Review the rules and guidelines to understand how to manage the `.csv` file and how the import job processes data from the `.csv` file.

Consider the following rules and guidelines when you import information from a `.csv` file:

- The first row of the `.csv` file must contain case-sensitive column headings in the required sequence. If the `.csv` file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines the imported information.
  For example, if you are importing users, each row defines a user. The combination of the UserName and UserDN columns defines a unique user. If the UserName and UserDN in one row is the same as the UserName and UserDN in a second row, the import job treats the second row as a duplicate of the first row.
- If the `.csv` file contains duplicate entries, the import job processes the entries in the order in which the rows appear in the file.
- The column values cannot contain the following keywords or terms:

```
SELECT FROM
SELECT UNION
DELETE FROM
```

```
UPDATE SET
<SCRIPT
</SCRIPT>
ALERT(
"/>"
```

If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.

- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.
- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.

Format to Import Users from a .csv File

Create a `.csv` file to create, update, or delete information about users in the Data Privacy Management repository. The content of the `.csv` file must be in the order and format that the import job requires. The `.csv` file must contain all of the column headings. However, not all columns require values.

The following table lists the order of the columns, column headings, and the format for the values in the `.csv` file:

Column Order	Column Heading	Column Value Description
1	UserName	Required. Name of a user account that accesses information in a data store. The combination of the UserName and UserDN columns identifies a unique user to create, update, or delete. If the UserDN is blank, the UserName must be unique. If blank, the import job rejects the row.
2	UserDN	Optional. Distinguished name for the principal user. The combination of the UserName and UserDN columns identifies a unique user to create, update, or delete. If blank, the UserName identifies a unique user.
3	FirstName	Optional. First name of the user.
4	LastName	Optional. Last name of the user.
5	FullName	Optional. Full name of the user. If blank, the import job populates the property with the values from the FirstName and LastName columns.
6	Email	Optional. Email address of the user.
7	Title	Optional. Job title of the user.
8	Department	Optional. Department of the user. This department is not related to the data store departments in the Data Privacy Management repository. If blank, default is UNKNOWN.
9	Location	Optional. Location of the user. This location is not related to the data store locations in the Data Privacy Management repository. If blank, default is UNKNOWN.

Column Order	Column Heading	Column Value Description
10	Manager	Optional. Manager of the user.
11	Organization	Optional. Organization of the user.
12	UserStatus	Optional. Status of the user. Enter one of the following values: - A. Active - I. Inactive If blank, default is A (active).
13	Action	Optional. Instructs the import job to create, update, or delete a user in the Data Privacy Management repository. Enter one of the following values: - U. Creates or updates a user in the Data Privacy Management repository. The action that the job performs depends on if the user exists in the Data Privacy Management repository and on the Replace Duplicates option. You configure the Replace Duplicates option when you import the The following table lists the order of the columns, column headings, and the format for the values in the `.csv` file: file. If the user does not exist, the import job creates the user. If the user exists and the Replace Duplicates option is enabled, the import job updates the user. **Warning:** If a column in the The following table lists the order of the columns, column headings, and the format for the values in the `.csv` file: file contains a null value, the import job replaces the value in the Data Privacy Management repository with a null value. If the user exists and the Replace Duplicates option is not enabled, the import job skips the row. - D. Deletes a user from the Data Privacy Management repository. The user activity data related to the user remains. If the user does not exist, the import job rejects the row. If blank, default is U. Creates or updates a user in the Data Privacy Management repository.

Format to Import User Groups from a .csv File

Create a `.csv` file to create, update, or delete information about user groups in the Data Privacy Management repository. The content of the `.csv` file must be in the order and format that the import job requires. The `.csv` file must contain all of the column headings. However, not all columns require values.

The following table lists the order of the columns, column headings, and the format for the values in the `.csv` file:

Column Order	Column Heading	Column Value Description
1	GroupName	Required. Name of a group. The combination of the GroupName and GroupDN columns identifies a unique group to create, update, or delete. If the GroupDN is blank, the GroupName must be unique. If blank, the import job rejects the row.
2	GroupDN	Optional. Distinguished name for the group. The combination of the GroupName and GroupDN columns identifies a unique group to create, update, or delete. If blank, the GroupName determines a unique group.
3	Description	Optional. Description of the group.
4	ManagedBy	Optional. Owner of the group.
5	GroupType	Optional. Type of group.
6	GroupScope	Optional. Scope of group.
7	Action	Optional. Instructs the import job to create, update, or delete a group in the Data Privacy Management repository. Enter one of the following values: - U. Creates or updates a group in the Data Privacy Management repository. The action that the job performs depends on if the group exists in the repository and on the Replace Duplicates option. You configure the Replace Duplicates option when you import the The following table lists the order of the columns, column headings, and the format for the values in the `.csv` file: file. If the group does not exist, the import job creates the group. If the group exists and the Replace Duplicates option is enabled, the import job updates the group. **Warning:** If a column in the The following table lists the order of the columns, column headings, and the format for the values in the `.csv` file: file contains a null value, the import job replaces the value in the Data Privacy Management repository with a null value. If the group exists and the Replace Duplicates option is not enabled, the import job skips the row. - D. Deletes a group and associated group memberships from the repository. If the group does not exist, the import job rejects the row. If blank, default is U. Creates or updates a group in the repository.

Format to Import Group Memberships from a .csv File

Create a `.csv` file to create, update, or delete information about group membership in the Data Privacy Management repository. The content of the file must be in the order and format that the import job requires. The file must contain all of the column headings. However, not all columns require values.

Import group membership after you import users and groups.

The following table lists the order of the columns, column headings, and the format for the values in the `.csv` file:

Column Order	Column Heading	Column Value Description
1	MemberName	Required. Name of the user or group that is a member of a group. The combination of the MemberName and MemberDN columns identifies a unique member for which to create, update, or delete a membership. If blank, the import job rejects the row.
2	MemberDN	Distinguished name of the member. The combination of the MemberName and MemberDN columns identifies a unique member for which to create, update, or delete a membership. Required if the member has a member distinguished name in the Data Privacy Management repository.
3	MemberType	Optional. Indicates the type of member to assign to a group. Enter one of the following values: - U. The member is a user. - G. The member is a group. If blank, default is U (User).
4	MemberOf	Required. Name of the group to which the member is assigned. If blank or the group does not exist in the repository, the import job skips the row.
5	MemberOfDN	Optional. Distinguished name of the parent group.
6	Action	Optional. Instructs the import job to create, update, or delete a group membership in the repository. Enter one of the following values: - U. Creates or updates a group membership in the repository. The action that the job performs depends on if the group membership exists in the repository and on the Replace Duplicates option. You configure the Replace Duplicates option when you import the .csv file. If the group membership does not exist, the import job creates the group membership. If the group membership exists and the Replace Duplicates option is enabled, the import job updates the group membership. If the group membership exists and the Replace Duplicates option is not enabled, the import job skips the row. - D. Deletes a group membership from the repository. If the group membership does not exist, the import job rejects the row. If blank, default is U. Creates or updates a group membership in the repository.

Format to Import User Aliases from a .csv File

Create a `.csv` file to create or delete information about user aliases in the Data Privacy Management repository. The content of the file must be in the order and format that the import job requires. The file must contain all of the column headings. However, not all columns require values.

The following table lists the order of the columns, column headings, and the format for the values in the `.csv` file:

Column Order	Column Heading	Column Value Description
1	UserName	Required. Name of a user account that accesses information in a data store. The combination of the UserName and UserDN columns identifies a unique user for which to create or delete an alias. If blank, the import job rejects the row.
2	UserDN	Distinguished name for the principal user. The combination of the UserName and UserDN columns identifies a unique user for which to create or delete an alias. Required if the user has a user distinguished name in the repository.
3	AliasName	Required. Alternative name or alias for a user account. You can specify one alias in a row. To specify multiple aliases for a user, create one row for each alias. If blank, the import job rejects the row.
4	Action	Optional. Instructs the import job to create or delete an alias in the repository. Enter one of the following values: - U. Creates an alias in the repository. - D. Deletes an alias from the repository. If blank, default is U. Creates an alias in the repository.

Format to Import User Access from a .csv File

Create a `.csv` file to create, update, or delete information about user access information in the Data Privacy Management repository. The content of the file must be in the order and format that the import job requires. The file must contain all of the column headings. However, not all columns require values.

The following table lists the order of the columns, column headings, and the format for the values in the `.csv` file:

The following table lists the order of the columns, column headings, and the format for the values in the `.csv` file:

Column Order	Column Heading	Column Value Description
1	MemberName	Required. User or group name for which you want to import access levels. The combination of the MemberName and MemberDN columns identifies a unique user to create, update, or delete. If blank, the import job rejects the row.
2	MemberDN	Distinguished name of the user or group. The combination of the MemberName and MemberDN columns identifies a unique user to create, update, or delete. Required if the member has a member distinguished name in the Data Privacy Management repository.
3	MemberType	Optional. Determines the type of member for the member name. Enter one of the following values: - U. The member name is a user. - G. The member name is a group. If blank, default is U (User).
4	AccessType	Optional. Determines the level of access that the user or group has to a data store. Enter one of the following values: - R. Read access. - W. Write access - D. Delete access. - RWD. Read, write, and delete access. - All. Includes all administrator privileges, such as modify schema and tables. If blank, default is All.
5	DataStore	Required. Name of the data store to which the user has access. You can specify one data store in a row. To specify access to multiple data stores for a member, create one row for each data store. If the data store name is blank or does not exist in the repository, the import job rejects the row. If you do not enter a SchemaName, TableName, or ColumnName, the import job sets access to all schemas and tables for the data store.
6	SchemaName	Optional. Name of the schema that identifies the table to which the user or group has access. The name is case sensitive. If you specify a SchemaName, you must specify a TableName. If blank, the import job sets access to all schemas that Data Privacy Management identified as sensitive for the data store. If the table is not identified as sensitive in the repository or if the case does not match, the import job rejects the row.
7	TableName	Required if you specified a SchemaName or a ColumnName. Name of the table in the data store to which the user or group has access. The name is case sensitive. If blank and you did not specify a SchemaName, the import job sets access to all tables that Data Privacy Management identified as sensitive for the data store. If blank and you did specify a SchemaName, the import job rejects the row. If the table is not identified as sensitive in the repository or if the case does not match, the import job rejects the row.

Column Order	Column Heading	Column Value Description
8	ColumnName	Optional. Name of the column to which the user or group has access. The name is case sensitive. If you specify a ColumnName, you must specify a TableName. If blank, the import job sets access to all columns that Data Privacy Management identified as sensitive for the table in the data store. If the column is not identified as sensitive in the repository or if the case does not match, the import job rejects the row.
9	Action	Optional. Instructs the import job to create, update, or delete user access in the repository. Enter one of the following values: - U. Creates or updates user access in the repository. The action that the job performs depends on if the user access exists in the repository and on the Replace Duplicates option. You configure the Replace Duplicates option when you import the `.csv` file. If the user access does not exist, the import job creates the user access. If the user access exists and the Replace Duplicates option is enabled, the import job updates the user access. **Warning:** If a column in the `.csv` file contains a null value, the import job replaces the value in the repository with a null value. If the user access exists and the Replace Duplicates option is not enabled, the import job skips the row. - D. Deletes user access from the repository. If the user access does not exist, the import job rejects the row. If blank, default is U. Creates or updates user access in the repository.

Importing User Information from a .csv File

You can import user information from a `.csv` file. The file must be in the format that the import job requires.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.

2. From the **Actions** menu, click **Import** and then click the required option for users, user groups, user group memberships, user access, or user alias information import.

   The corresponding **Import** page appears.

3. Enable or disable the **Replace Duplicates with Items Imported** option.

   The option determines how the import job handles updates for users that exist in the Data Privacy Management repository.
   If you enable the option, the import job updates the user properties in the Data Privacy Management repository with the values in the .csv file.

   Warning: If a column in the `.csv` file contains a null value, the import job replaces the value in the Data Privacy Management repository with a null value.

   If you disable the option, the import job skips the row.

4. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

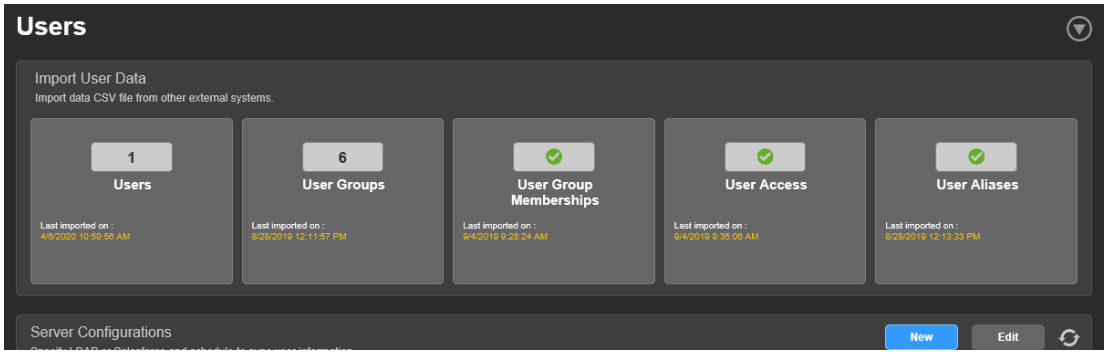
If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

5. Click **Import**.

   The Import job begins. To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the `.csv` file, you can download a list of the rejected rows.

When the job completes, you can view the imported user information from the **Users** workspace. Click **Manage** > **Users** to open the Users workspace.

The following image shows the summary data in the **Users** workspace:



Event Formats

Data Privacy Management supports user activity event messages in CEF, LEEF, and JSON formats.

The following file contains sample formats with the required fields for supported event message types: `<Informatica installation directory>/SecureAtSourceService/config/EventFormats.conf`

The following sections contain examples of event formats. You can modify the values for your environment.

Note: The properties in **BOLD** indicate the required properties for each message format.

CEF Event Message Format

```
CEF: 0|Any DAM Tool|XXX|10.5.0.4_0|Audit|Audit.DAM|Informative|dst=10.0.0.1 dpt=0 duser=<user
name> src=10.77.3.1 spt=0 proto=TCP rt=17 August 2020 9:00:10, cat=Audit cs1Label=Policy
cs2=Database Server Group cs2Label=ServerGroup cs3=DB Services cs3Label=ServiceName cs4=Default
cs4Label=ApplicationName cs5=1 cs5Label=EventId cs6=Query cs6Label=EventType cs7=Default
cs7Label=UserGroup cs8=True cs8Label=UserAuthenticated cs9= cs9Label=ApplicationUser cs10=jdbc
thin client cs10Label=SourceApplication cs11=any cs11Label=OSUser cs12=server
cs12Label=HostName cs13=<Database name> cs13Label=Database cs14=<schema name> cs14Label=Schema
cs15= cs15Label=RawQuery cs16=<Query> cs16Label=ParsedQuery cs17= cs17Label=BindVariables cs18=
cs18Label=SQLError cs19=0 cs19Label=ResponseSize cs20=57 cs20Label=ResponseTime cs21=<Number of
affected rows> cs21Label=AffectedRows
```

LEEF Message Format

```
LEEF:1.0|Imperva Inc.|SecureSphere| 10.5.0.4_0| Logout| Server Group=Database Server Group|
Service Name=Oracle Services| ApplicationName=Default Oracle Application| Source Type=Agent|
User Type=Valid| usrName=<user name>| UserGroup=Default| Authenticated=True| App User=|
src=10.77.3.1| Application=jdbc thin client| OS User=oracle| Host=pbox-montreal| Service
Type=Oracle| dst=10.0.0.1| Event Type=Logout| Operation=Logout| Operation type=| Object name=$
{Event.struct.operations.objects.name}| Object type=${Event.struct.operations.objectType}|
Subject=${Event.struct.operations.subjects.name}| Database=SR_BANK_HR| Schema=SR_BANK_HR| Table
Group=| Sensitive=False| Privileged=False| Stored Proc=False| Completed Successfully=True| Raw
Data=| Parsed Query=SELECT * FROM SR_BANK_HR.EMPLOYEES| Bind Vaiables=| Error=| Response
Size=0| Response Time=0| Affected Rows=19| devTimeFormat=[yyyy-MM-dd HH:mm:ss.S]| devTime=28
August 2020 17:19:40
```

JSON Message Format

```
15/03/27 21:52:40 2015-03-27 21:52:40,791 INFO provider.Log4jAuditProvider: {
"resource": "default/sample_07/description, salary",
"resType": "@column",
"reqData": "SELECT \"FIRSTNAME\", \"LASTNAME\", \"SSN\", \"EMAIL\", \"DOB\" FROM
\"SR_BANK_PAYROLL\".\"EMPLOYEE\"",
"reqUser": "<user name>",
"evtTime": "20200808-03:19:05.395-+0000",
"policy": 0,
"access": "SELECT",
"result": 81,
"enforcer": "xasecure-acl",
"repoType": 3,
"repo": "SR_BANK_PAYROLL",
"sess": "efde5a24-2ef8-4994-b4d1-4fd7ac42ba04",
"cliType": "HIVESERVER2",
"cliIP": "10.77.3.1",
"action": "QUERY"
}
```

Configuring Event Streaming with Informatica Database AuditTrail Utility

You can use the Informatica Database AuditTrail Utility to load user activity information from an Oracle, IBM DB2, or Microsoft SQL Server database.

Perform the following high-level tasks to configure event streaming with Informatica Database AuditTrail Utility:

1. Enable database audit on the database that you want to monitor.
2. Download and extract the Informatica Database AuditTrail Utility.
3. Update the `dbs.csv` file.
4. Configure the utility to stream events.

Configure the Database for Audit

Enable the database that you want to monitor for audit before you configure event streaming to Data Privacy Management.

▶ Run the required command based on the database.

- In an Oracle database, set `AUDIT_TRAIL=DB, EXTENDED`.
  `AUDIT_TRAIL=DB` directs audit records to the `SYS.AUD$` table, except for records that are always written to the operating system audit trail.

  `AUDIT_TRAIL=DB, EXTENDED` performs all actions of the `AUDIT_TRAIL=DB` setting and populates the SQL bind and SQL text CLOB-type columns of the `SYS.AUD$` table when available.

  For example: `ALTER SYSTEM SET AUDIT_TRAIL=DB, EXTENDED SCOPE=SPFILE;`
- In a Microsoft SQL Server database, run the commands for audit.
  The following code is an example of the commands:

```
USE master;
GO
GRANT ALTER ANY SERVER AUDIT TO AuditConfigurationLogin (audit user);
GO
GRANT ALTER ANY SERVER AUDIT TO sats_audit;
GO
-- Create the server audit.
CREATE SERVER AUDIT Security_Audit
    TO FILE (FILEPATH ='D:\MicrosoftSQLServer\Audit');
```

```
GO
-- Enable the server audit.
ALTER SERVER AUDIT Security_Audit WITH (STATE = ON);
USE <MY_DATABASE>;
GO
-- Create the database audit specification.
CREATE DATABASE AUDIT SPECIFICATION DB_AUDIT
FOR SERVER AUDIT Security_Audit  ADD (SELECT ON DATABASE::idvtest2 BY PUBLIC);
GO

CREATE SERVER AUDIT ACCESS_Audit  TO FILE (FILEPATH ='D:\MicrosoftSQLServer\Audit');
GO
CREATE SERVER AUDIT ACCESS_Audit ACCESS_Audit_Specification
FOR SERVER AUDIT HIPPA_Audit  ADD (FAILED_LOGIN_GROUP);
GO
CREATE SERVER AUDIT SPECIFICATION ACCESS_Audit_Specification
FOR SERVER AUDIT ACCESS_Audit
    ADD (FAILED_LOGIN_GROUP);
GO
-- Enables the audit.
ALTER SERVER AUDIT ACCESS_Audit
WITH (STATE = OFF);
```

- In an IBM DB2 database, run the commands for audit.
  The following code is an example of the commands:

```
db2audit configure datapath /home/bigsql/auditlogs archivepath /home/bigsql/archive

db2 "CREATE AUDIT POLICY LOGALL CATEGORIES ALL STATUS BOTH ERROR TYPE
AUDIT"

db2 "AUDIT DATABASE USING POLICY LOGALL"
```

Download and Extract the Utility Files

Use the Informatica Database AuditTrail utility to collect and stream user activity events.

The utility is available for download from tsftp.informatica.com at the following path: `\updates\ILM\SecureAtSource\Tools\UBA\DPM_DB_Audit.zip`

For any issues with access and download, contact Informatica Global Customer Support.

Extract the utility files to the following location:

`<Informatica installation directory>\DB_audit`

Create a `DB_audit` folder if it does not exist.

The `DPM_DB_Audit.zip` file contains the following files:

- `DatabaseAuditTool-4.1.0-SNAPSHOT.jar`. The database user activity event loader.
- `runDatabaseAuditTool.sh`. The shell script to manage the database user activity event loader.
- `dbs.csv`. A sample database user activity event loader input `.csv` file.
- `dbs2.csv`. A sample database user activity event loader input `.csv` file.
- `dataStoreAliases.csv`. A sample data store alias `.csv` file.
- `userAliases.csv`. A sample user alias `.csv` file.
- `Usermanagement_GroupMembership.csv`. A sample user group membership `.csv` file.
- `Usermanagement_Groups.csv`. A sample user group `.csv` file.
- `Usermanagement_Users.csv`. A sample users `.csv` file.

Update the dbs.csv File

The `dbs.csv` file that you download with the utility contains sample information. Replace the sample information with information specific to your setup.

The file contains columns with the following headers:

Header	Description
dbType	The type of database from which you stream event data.
listenerHostname	Host name of the Data Privacy Management machine.
listenerPort	Port number on which the Data Privacy Management TCPListener listens. Default is 51000.
dbHost	Host name of the database from which you stream event data.
dbPort	Port number of the database from which you stream event data.
dbName	Name of the database from which you stream event data.
dbUser	The database user name.
dbPassword	The database password.
optionalField	Required for Microsoft SQL Server databases. Enter the user audit file name with the complete path. For example: `C:\Program Files\Microsoft SQL Server\MSSQL10_50.PC_SQLS\MSSQL\DATA\*.sqlaudit`
excludeUsers	User names that you want to exclude from audit collection. For example, you might want to exclude database system users. Use commas to separate multiple values.

Enter the required values in each column and save the changes to the file. Do not update the column headers or the file name.

Configure the Utility to Stream Events

Configure the utility to fetch user activity events, generate message events in the required format, and stream the events to the Data Privacy Management listener.

1. Run the following command to set Execute permission on the shell script file:

   ```
   chmod +x ./runDatabaseAuditTool.sh
   ```

2. Run the following command to fetch all collected user activity events, generate CEF message events, and stream the events to the Data Privacy Management listener.

   ```
   ./runDatabaseAuditTool.sh DEBUG ./dbs.csv
   ```

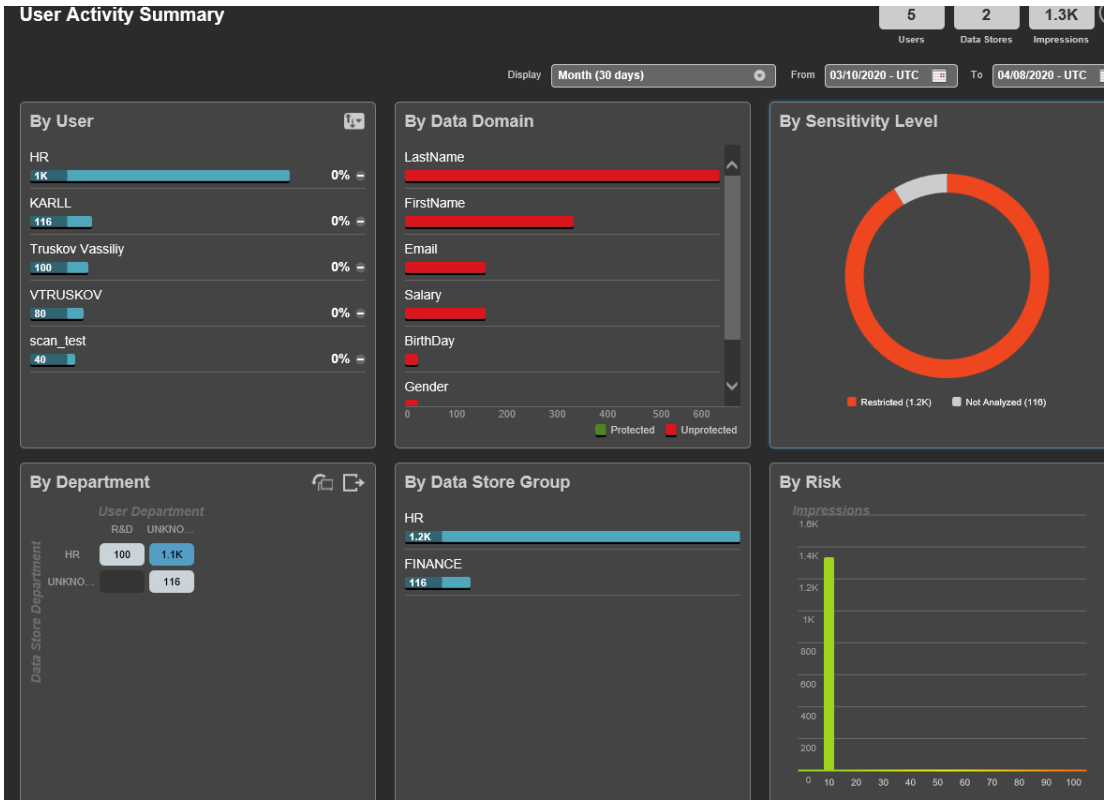
   When the script runs, it generates a `checkpoint_<DataStoreAliasname>` checkpoint file in the local folder. For example: `checkpoint_hrdb_server.informatica.com_HRDB_1433`

3. To automate the load process, add the command to the crontab list for the user. To load events every hour, run the following command:

   ```
   crontab -e <Informatica installation directory>/DB_audit/runDatabaseAuditTool.sh
   DEBUG ./dbs.csv >> <Informatica installation directory>/DB_audit/ua_load.dat 2>&1
   ```

4. Validate the added process with the crontab -l command.

After the messages stream to the application, you can log in to Data Privacy Management and view user activity event information on the **User Activity Summary** workspace.
The following image shows a sample **User Activity Summary** workspace:



To view anomalies and violations, you must create Security Policies. See the *Informatica Data Privacy Management User Guide* for information on how to configure violation and anomaly detection.

Author

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